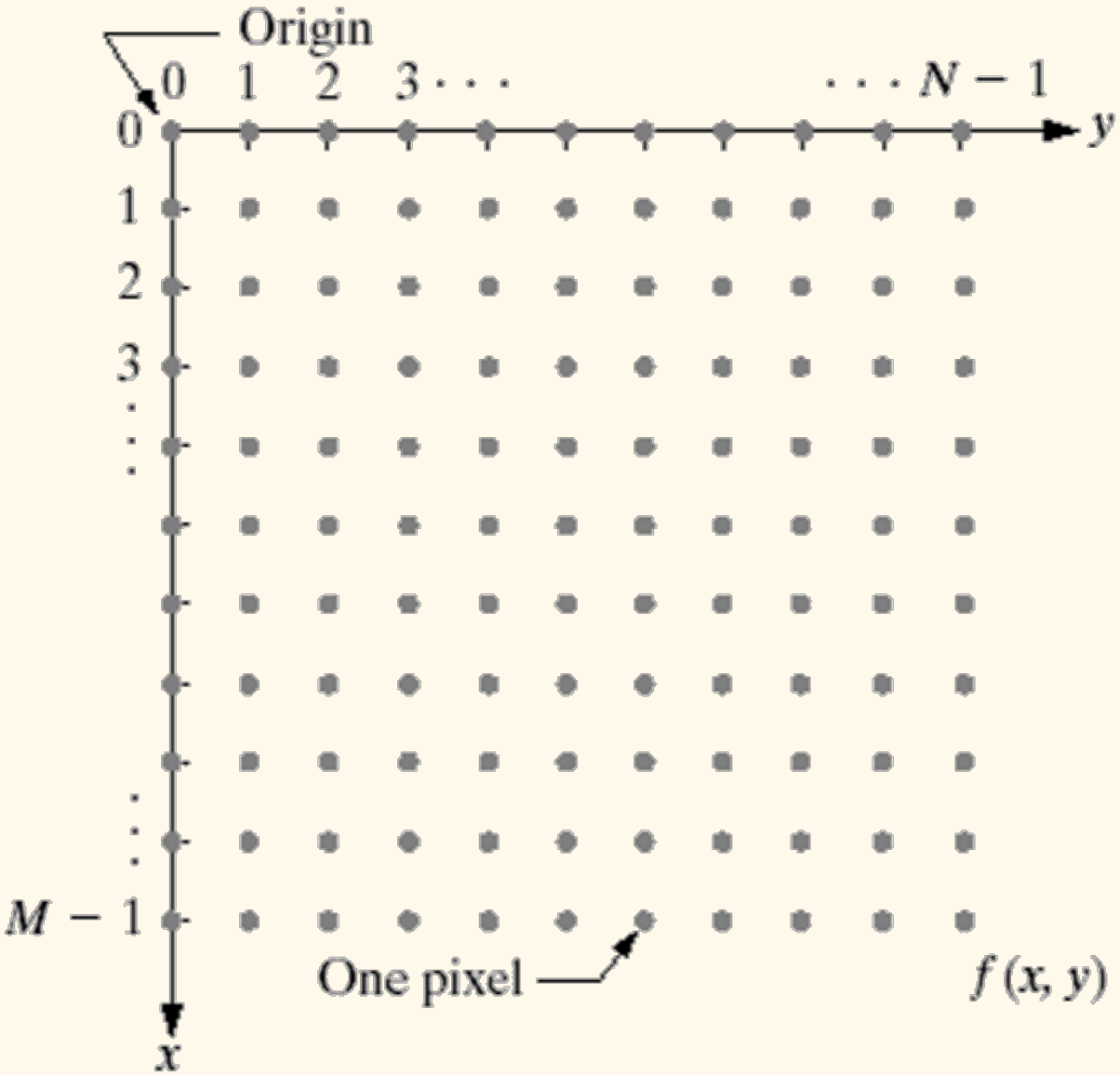
Image processing: basic definitions and related domains

What is digital image processing?

A gray level digital image



Context

Image processing

Image analysis

Image understanding

Computer vision

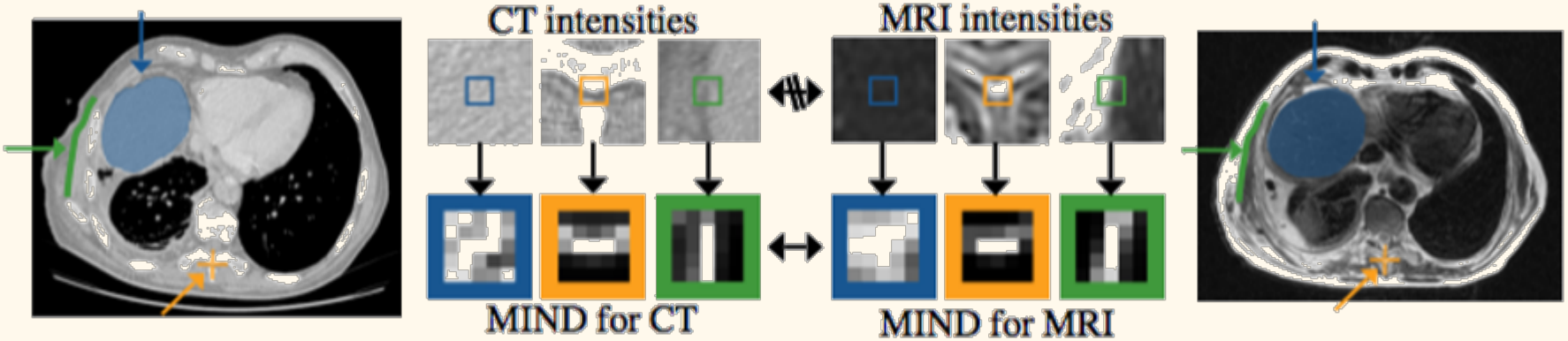
What are example operations?

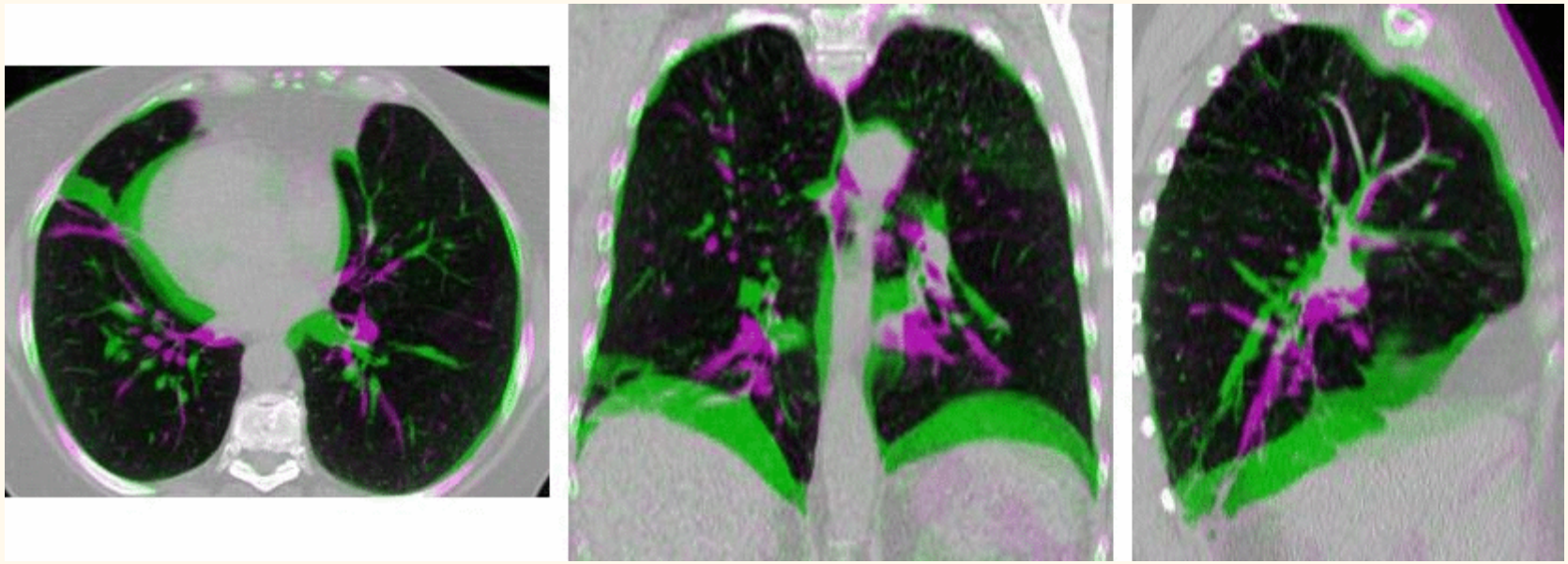
Image enhancement

Image enhancement

Image registration and tracking

Image registration and tracking





Credit: Integrated Analysis and Probabilistic Registration of Medical Images with Missing Correspondences
M.Sc. Sandra Schultz
Dr. rer. nat. Jan Ehrhardt
Prof. Dr. rer. nat. habil. Heinz Handels

Image registration and tracking

Image is public domain

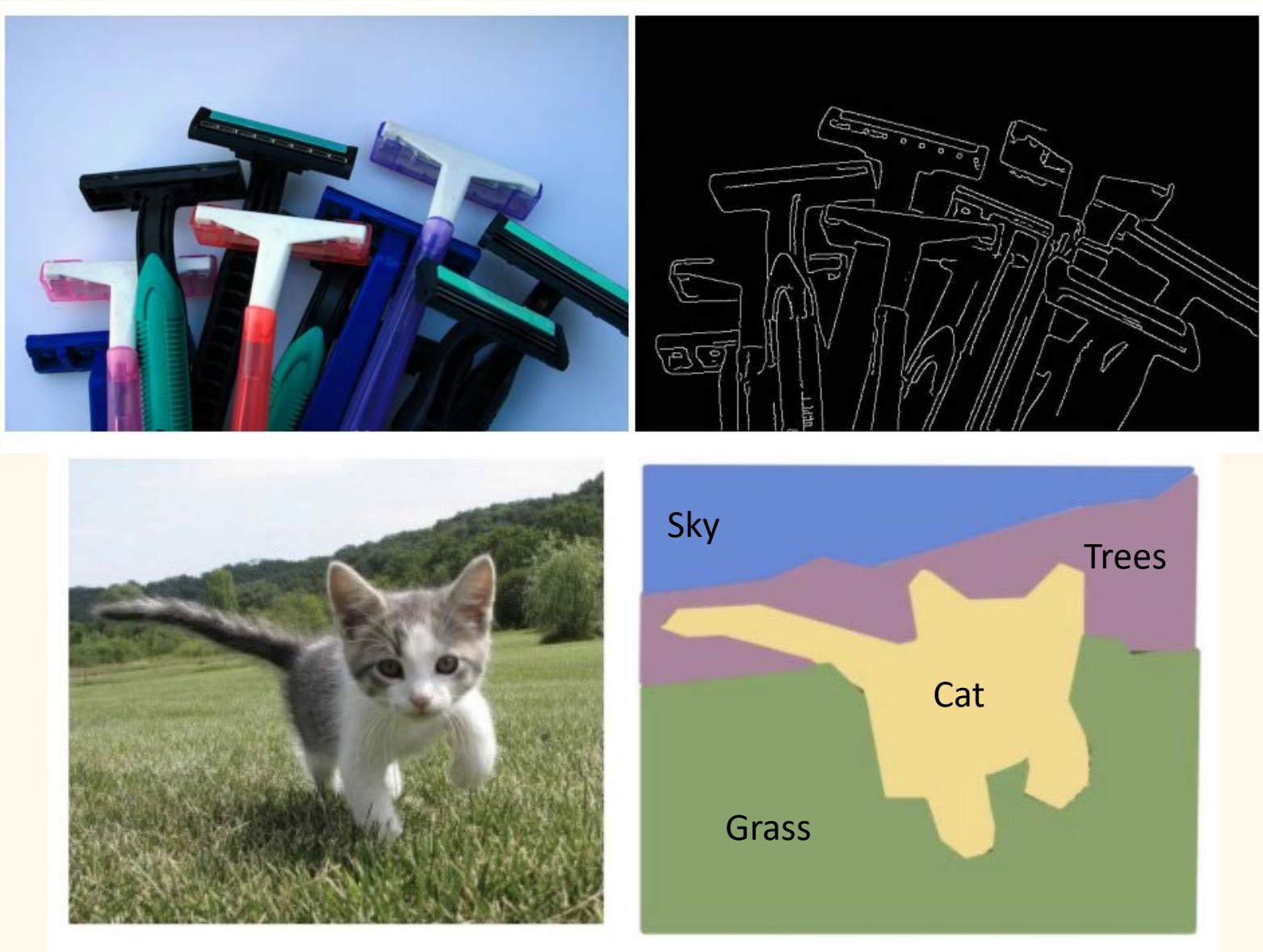
Image is public domain

"SIFT" & Object Recognition, David Lowe, 1999

Credit: Fei-Fei Li, Justin Johnson, Serena Yeung

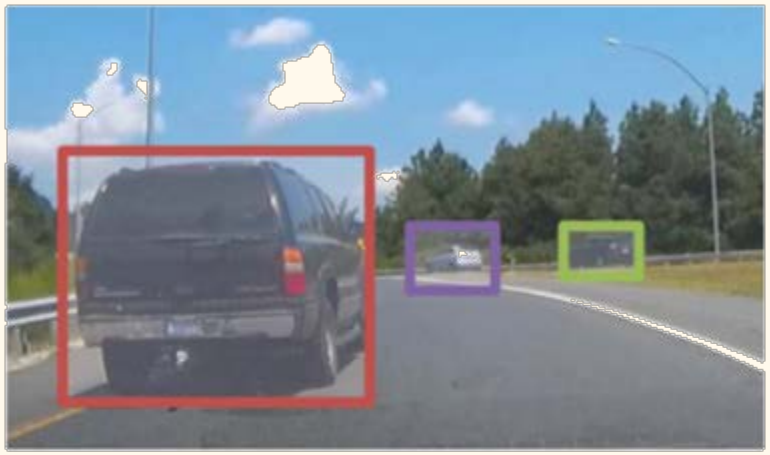
Image segmentation

Image segmentation

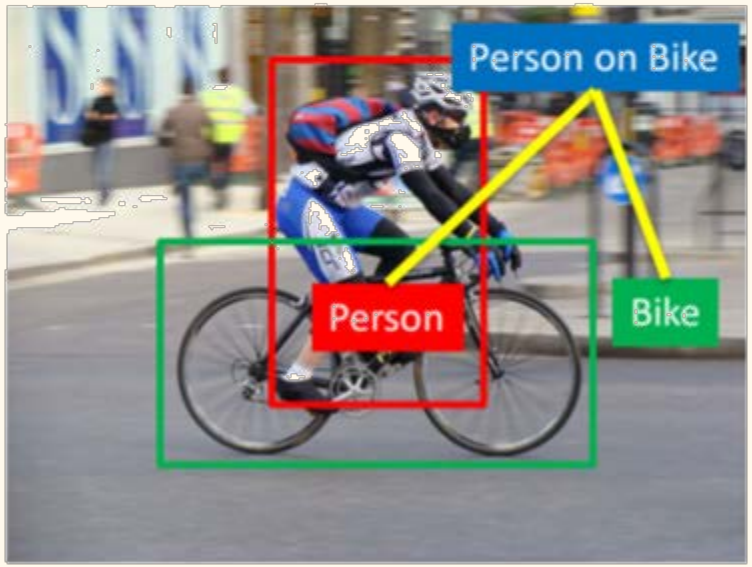


Credit: Fei-Fei Li, Justin Johnson, Serena Yeung

More intelligent operations



- Object detection
- Action classification
- Image captioning
- …



Credit: Fei-Fei Li, Justin Johnson, Serena Yeung

More intelligent operations

Context

Image processing

Image analysis

Image understanding

Computer vision

Context

Image: noisy with poor contrast

Image processing

Example operation: enhancement

Image: enhanced

Image analysis

Example operation : segmentation

Information: features such as diameters, areas, perimeters, numbers, coordinates

Image understanding

Example operation : classification, recognition

Understanding: Identification of objects

Computer vision

Example operation : Reconstruction

Visualization of objects: when occluded or from other angles

Context

Gonzalez Text Book:

Image: noisy with poor contrast

Low level processing

Example operation: enhancement

Image: enhanced

Mid level processing

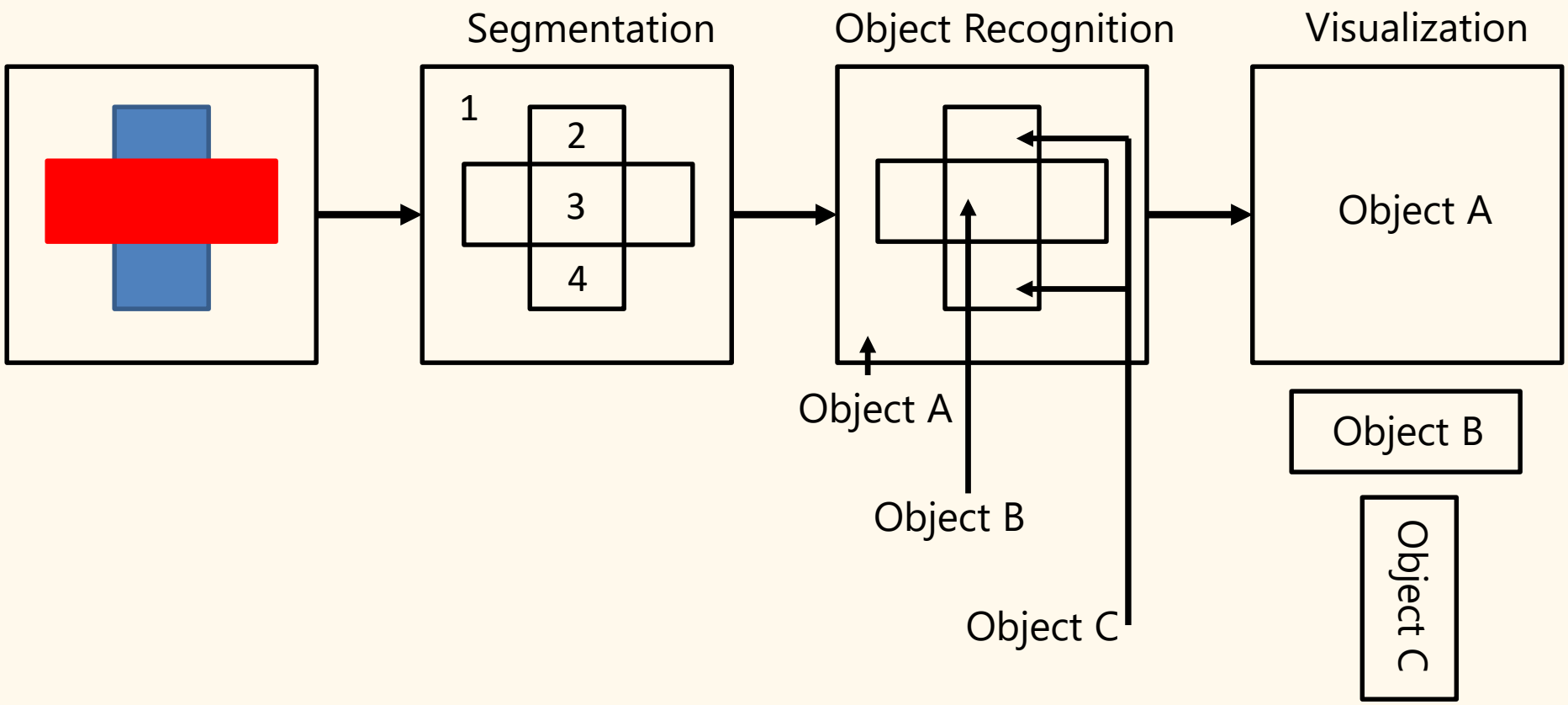
Example operation : segmentation

Information: features, contours, edges, diameters, areas, numbers, coordinates, classification, recognition

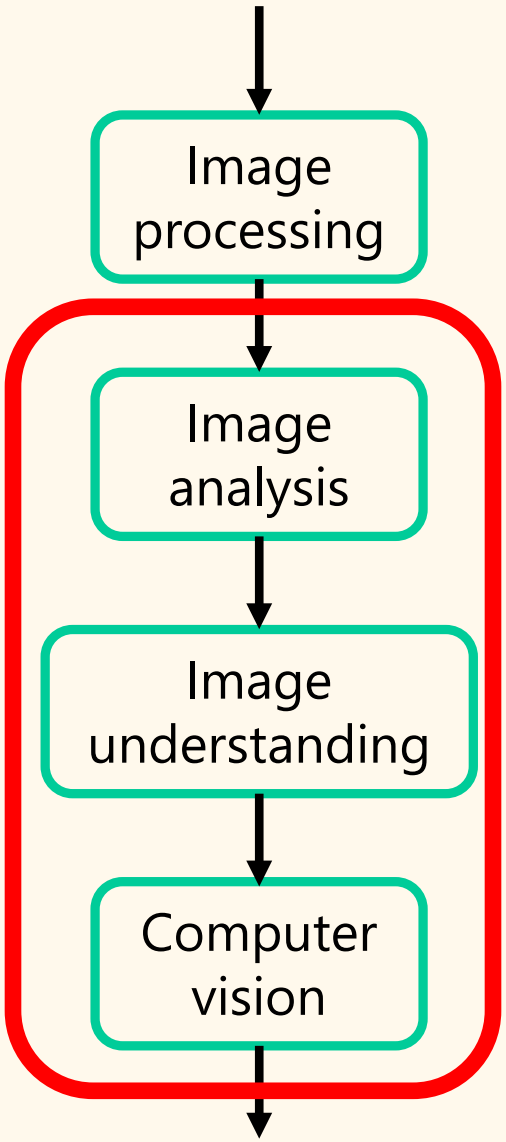
Image analysis and Computer vision

Making sense of the ensemble of objects

Context


Segmentation
Object Recognition
Visualization
1
2
3
4
Object A
Object B
Object C
Object A
Object B
Object C

Context



Lots of discrepancies in the literature! These 3 may overlap or be used in place of one another.

Final output could guide intelligent or cognitive functions related to vision

Context

Image processing

Image analysis

Image understanding

Computer vision (CV)

Machine vision

Speech recognition

Natural language processing (NLP)

Artificial intelligence

Machine learning (ML)

Deep learning (DL)

Why image processing?

Why image processing?

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Credit: Fei-Fei Li, Justin Johnson, Serena Yeung